Jinil Kim

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RESEARCH INTEREST

- Developing computational frameworks that extract latent signatures from human brain and behavior
- Translating these into real-world systems for personalized interventions and human-inspired AI
- Uncovering shared and distinct principles across brains, machines, and languages

EDUCATION

Seoul National University — Seoul, South Korea
B.A. Linguistics · B.A. Psychology · B.S. Computer Science — Mar 2021 – Exp. Feb 2027
GPA: 4.18 / 4.3 (3.97 / 4.0)
* Mandatory military service, Jul 2023 – Jan 2025

The University of Hong Kong — Hong Kong SAR
Summer Research Program, Faculty of Psychology — Jun 2025 – Aug 2025

PUBLICATIONS & MANUSCRIPTS

Sole first author on all listed work.

Workshop Papers

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. (2026). *Inferring Individualized Color-Vision Distortions from fMRI Hue-Representation Geometry.* ICML 2026 Workshop on Structured Data for Health (SD4H), Seoul, Korea.

In Preparation

- **Kim, J.**, Cho, A. M., Seo, J., & Cha, J. *Individual-specific distortion of cortical hue geometry in color vision deficiency informs personalized color correction.*
- **Kim, J.**, & Yao, Y.-W. *When, not just how strongly: a delayed onset of attribute integration in depression.*
- **Kim, J.**, et al. *Inverse reinforcement learning of latent human planning.*

PRESENTATIONS

- **Kim, J.**, Cho, M., Seo, J., & Cha, J. (2026). *fMRI Decoding Reveals Intact Neural Color Representations in Color Vision Deficiency.* OHBM 2026 Annual Meeting, Bordeaux, France. [Poster]
- **Kim, J.**, & Yao, Y.-W. (2025). *When, not just how strongly: a delayed onset of attribute integration in depression.* HKU Summer Research Program, Hong Kong. [Poster]

RESEARCH PROJECTS

Individual-specific distortion of cortical hue geometry in color vision deficiency informs personalized color correction
Project Lead, sole first author (Advisor: Prof. Jiook Cha) — Apr 2025 – Present

- Characterized individual-specific distortions of cortical hue geometry in color vision deficiency from fMRI, and translated them into personalized color-correction filters.
- Funded by the SNU Undergraduate Research Grant ($3,500).

When, not just how strongly: a delayed onset of attribute integration in depression
Project Lead, sole first author (Advisor: Prof. Yuan-Wei Yao) — Jun 2025 – Present

- Built a time-varying drift-diffusion model with an onset-latency parameter (τ) showing that depression integrates the second decision attribute later in social decisions.

Inverse Reinforcement Learning of Latent Human Planning
Project Lead, sole first author (Advisor: Prof. Woo-Young Ahn) — Sep 2025 – Present

- Developing an inverse reinforcement learning framework that more faithfully models human decision-making while recovering latent planning depth from behavior alone.

RESEARCH EXPERIENCE

Connectome Lab — Seoul National University
Undergraduate RA (Advisor: Prof. Jiook Cha) — Jul 2026 – Present
Research area: Brain foundation models; neural fields

Computational Clinical Science Lab — Seoul National University
Undergraduate RA (Advisor: Prof. Woo-Young Ahn) — Apr 2025 – Present
Research area: Inverse RL; computational psychiatry; multimodal digital phenotyping

Mental health, Internet, Neuroscience & Decision-making (MIND) Lab — University of Hong Kong
Visiting Researcher (Advisor: Prof. Yuan-Wei Yao) — Jun 2025 – Aug 2025
Research area: Computational modeling of social decision-making

SCHOLARSHIPS & FELLOWSHIPS

KOSAF Humanity 100 Years Undergraduate Scholarship — Mar 2025 – Jul 2026
Amount: $12,500

SNU Undergraduate Research Grant — Apr 2025
Amount: $3,500

The University of Hong Kong Foundation for Educational Development and Research Scholarship — Jun 2025 – Aug 2025
Amount: $1,300

Superior Academic Performance Scholarship — Mar 2022, 2023; Sep 2022
Amount: $1,650 each

HONORS & AWARDS

Best Poster Award, SNU Undergraduate Research Program — Jan 2026
SNU Liberal Arts Competition (1st Place) — Jul 2023

TEACHING

Teaching Assistant, Basic Computing — Seoul National University
English (Mar 2026) · Korean (Sep 2025)

LANGUAGES

Human: Korean (Native), English (Full Professional Proficiency), French (Beginner)
Programming: Python, R, Stan, C++, Java